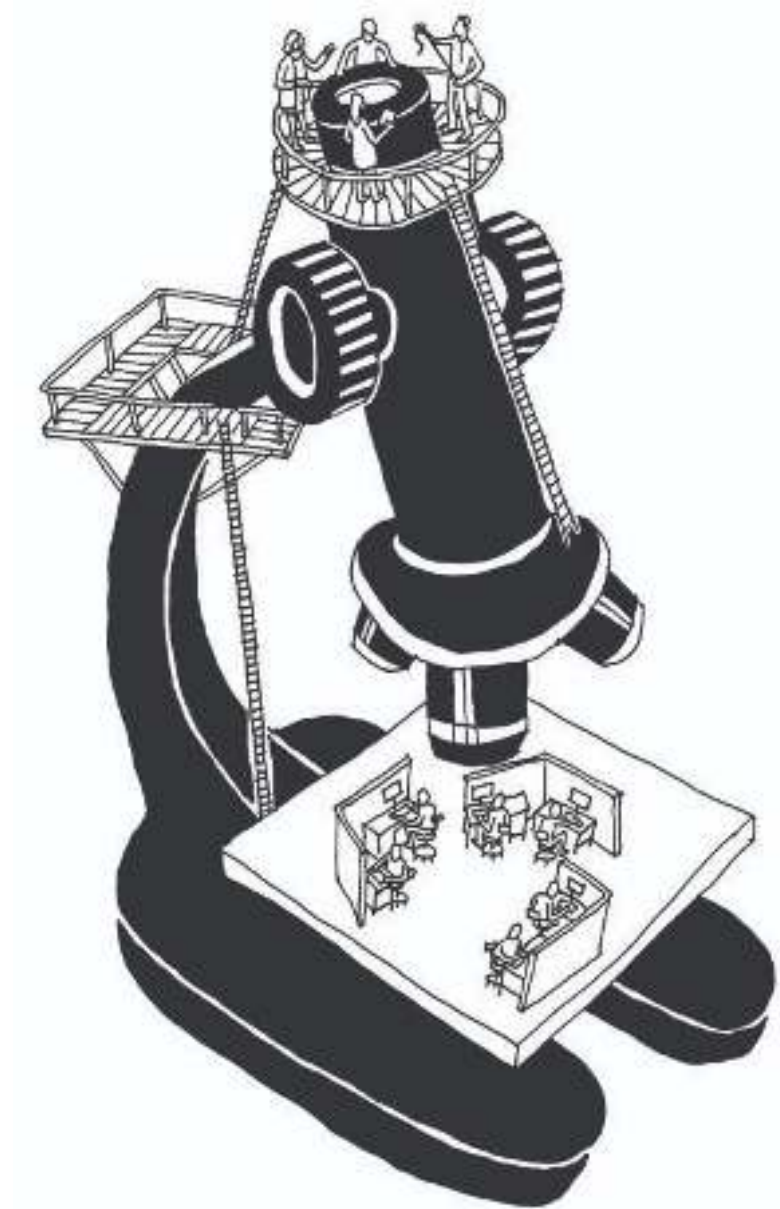


ECS 260: Software Engineering

How to read research papers

based on

“How to read engineering research papers” by William Griswold



How To Read a Research Paper

Papers are written in a **very condensed style** because of:

- **page limitations** and
- **the intended audience**, which is assumed to already know the area well.

The **reasons for writing the paper are different than the reasons the paper has been assigned**. So, **you have to work harder to find the content that you are interested in**.

Finally, **your time is very limited**, so you may not have time to read every word of the paper or read it several times to extract all the nuances.

For all these reasons, reading a research paper can **require a special approach**.

Standard Paper Organization

- Abstract
- Introduction
- Background/Theory
- Research Questions
- Methodology
- Results
- Discussion
- Threats to validity
- Conclusion
- Bibliography

Standard Paper Organization

- Abstract: summary, rarely read
- Introduction: motivation, why study this, summary of findings
- Background/Theory: where do your questions connect to prior
- Research Questions: well developed, precise questions
- Methodology: data, methods of inquiry, stats, etc
- Results: outcomes from methods application
- Discussion: interpret the results, connect to theory
- Threats to validity: weakness of approaches
- Conclusion: recap of findings, broader impact
- Bibliography: must be pertinent, current, complete

Two Things to Know

- (1) What you should get out of the paper
 - Are you a reviewer? Evaluate
 - Are you a practitioner? Actionables
 - Are you working in the area? What are the new ideas?
 - Student? Understanding and critical reading
- (2) Where is that information located in the paper
 - Last slide!
 - Some repetitions exist: intro, results/discussions, conclusion
 - Can read out of order (yes you can!)

Questions That Reading a Paper Should Answer

- What are the motivations for this work? What's the problem?
- What is the proposed solution?
- How is the solution evaluated?
- What is your take on the problem, solution, evaluation?
- What are the contributions? Ideas, artifacts, data.
- Future directions?
- What questions are you left with?
- Takeaway message?

Questions That Reading a Paper Should Answer

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Where to find?



- Abstract
- Introduction
- Background/Theory
- Research Questions
- Methodology
- Results
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Be Organized and Disciplined

- Take notes
 - When you get experienced on the paper is fine
- Have a standard approach
 - E.g., using the 8 questions above
- Keep a bibliography database with abstract and your notes (EndNote, bibtex, etc)
 - Enter your thoughts right after you read the paper